

GAUNTLET

The infrastructure crash-test rig for real-time voice AI.

Synthetic callers · acoustic measurement · seeded chaos · deterministic score · CI gate



Voice agents fail on timing — and nobody measures timing

What breaks in production

- The right answer, 1.9 s late
- Still talking after the caller interrupts
- Dead air that makes callers repeat themselves
- Fine at 1 call, broken at 20

Why today's tools miss it

- APM traces stop at the service boundary
- Transcript evals have no timing
- HTTP load tests have no media or conversation state
- One developer, one quiet call is not a test

A streaming inference chain under a perceptual deadline

- **Stateful:** failures appear at turn five, not turn one
- **Perceptual latency:** silence → endpointing → STT → LLM → TTS → first audio
- **Continuous media:** impairment shows up as jitter and loss, not a 500
- **Positive feedback:** late answers make callers repeat, which makes answers later

Latency, reliability, quality and cost trade against each other — shorten endpointing and you interrupt callers; add retries and you add latency. They only mean something when measured together, in the same run, against explicit thresholds.

What GAUNTLET does



Black-box: no code access, no instrumentation of the agent. One command in CI: `uses: .../gauntlet-gate`

One process, both directions, one clock

METRIC	DEFINITION
Response latency p50/p95/p99	caller's last voiced sample → agent's first audio
Barge-in stop time p95	interruption onset → agent stops; non-yield censored at 2 s
Talk-over · dead air	simultaneous speech · agent-side silences > 1.5 s
Completion · task success	no transport fault · goal met with verbatim turn citations
Degradation ratio	impaired p95 ÷ clean p95, same target and suite

Energy detector with an adaptive floor and sub-frame onset timing; high-resolution clock on every OS.

Reproducible adverse conditions

Frame loss

Gilbert bursts, faded substitution

Jitter · delay

truncated-normal release times, fixed delay subtracted from latency

Noise

café / street beds at a target SNR

Barge-in

interrupt at onset + seeded offset

Hesitant callers

mid-sentence pauses, slower speech

Mid-run switch

change conditions live; results split by epoch

Every stage draws from its own seeded stream: **same seed** → **identical schedule**.

A measurement tool publishes its own error

3.897 ms

MEASURED measurement error bound, 80 loopback repetitions at 200 / 500 / 1000 / 2000 ms. CI fails above

THRESHOLD 50 ms.

Loopback probe-and-pipeline bound — not a wide-area claim.
calibration/results.csv

8 calls

MEASURED sustained concurrently in one process on a laptop, timing error p95 $\approx$ 7 ms; saturates at 12. Aim of 10 **TARGET** not yet met.

benchmarks/rig-2026-09-11.csv

RESULT

Same suite, same seed, two agent tunings



Tuned · mobile network

score 100.0 · latency p95 786 ms · dead air 0 %



Slow endpointing · mobile

score 76.17 · latency p95 1,639.5 ms · dead air 15.6 %

✗ Gate failed — 4 breaches

readiness -23.83 · latency p95 +853 ms · time to first response +849 ms · dead air +15.6 pts

MEASURED

24 calls each on a laptop; bundled synthetic agent in scripted mode without provider keys, so task success unscored. benchmarks/runs-2026-09-11/

Models only where judgement is unavoidable

Uses a language model

- The caller's *words*
- Task-success judgement — verbatim citations, two passes, disagreement → needs-review

Deterministic code

- When to speak, interrupt, hang up
- Condition injection
- Measurement and scoring
- Gate and cost

A benchmark whose conditions a model chooses is not reproducible; a score a model assigns cannot be argued with.

Narrow on purpose

	VOICE-EVAL PLATFORMS	OBSERVABILITY	HTTP LOAD TOOLS	GAUNTLET
Conversational load	yes	no	no	yes
Acoustic latency, barge-in as metrics	not emphasised	no	no	primary
Seeded media impairment	not emphasised	no	network-level	yes, disclosed scope
Published error bound	not common	—	—	yes
Telephony, production monitoring	yes	yes	—	not built

Revenue follows synthetic call minutes

Free

60 call-minutes / month, 1 target, public reports

Team · \$99/mo

1,000 minutes, 5 targets, CI gate, private reports

Usage

metered minutes beyond plan

Enterprise

self-hosted runner, retention, custom profiles

Buyer: the team that owns a voice agent in production. Trigger: a launch or a provider change.

No margin is claimed — rig cost has not yet been measured with real provider prices.

What is true today

Limits

- Impairment is application-layer, caller-outbound only
- Frame substitution approximates packet loss
- Error bound is a loopback bound
- Groq, ElevenLabs and Speechmatics verified on a live call; LiveKit implemented, not yet verified
- No hardware acceleration claimed

Next

- A run against a third-party agent; LiveKit verified
- Concurrency ramp and matrix sweeps
- SIP telephony, bidirectional impairment
- A public, reproducible voice-pipeline benchmark